

Thanh Duy Huynh

AI STUDENT · COMPUTER VISION · APPLIED AI & INTELLIGENT SYSTEMS

Can Tho, Vietnam | Open to relocation

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Computer Vision • Deep Learning • Video Understanding • Edge AI • Robotics



Profile

AI undergraduate with hands-on experience in computer vision, video understanding, object detection, person re-identification, and deployable deep learning systems. Experienced across research and production-oriented workflows, including dataset preparation, model training and evaluation, real-time video pipelines, ONNX/Docker deployment, and technical reporting. Contributed to peer-reviewed research and AI/robotics projects, with a strong interest in building reliable perception and intelligent systems for real-world applications.

Technical Skills

Programming & Data	Python, C/C++ fundamentals, NumPy, Pandas, data preprocessing, statistics, model evaluation
Deep Learning	PyTorch, TensorFlow/Keras, CNNs, attention mechanisms, LSTM, temporal modeling
Computer Vision	OpenCV, YOLO-based detection, image/video preprocessing, tracking, pose estimation, person re-identification, video understanding
Deployment & Tools	ONNX, TorchScript, basic TensorRT, Docker, REST API integration, Linux, Git, Jupyter, Google Colab
Robotics & Edge AI	Real-time camera pipelines, edge inference, sensor integration, ESP32, basic ROS/ROS2

Publications & Research

Similarity-Driven Grad-CAM++ for Explainable Person Re-Identification

ICDSAIA 2026

ACCEPTED CONFERENCE PAPER

2026

- Proposed a similarity-driven Grad-CAM++ framework for person re-identification, generating attribution maps from query-gallery similarity scores rather than conventional class logits.
- Evaluated explanation behavior on Market-1501 and MSMT17 using ResNet50 and MobileNetV2, covering true-positive and false-positive retrieval cases.
- Designed quantitative sensitivity analysis to measure person/background attribution distribution and explanation robustness under image perturbations.
- Accepted for presentation at ICDSAIA 2026. Paper ID: 111.

Attention Placement and Parameter Sharing in Omni-Scale Networks for Person Re-Identification: A Controlled Study

ICDSAIA 2026

ACCEPTED CONFERENCE PAPER

2026

- Conducted a controlled study of attention placement in OSNet, comparing Post, SharedBranch, and IndBranch strategies across multi-scale branches.
- Evaluated Aggregation Gate (AG), ECA, CBAM, and Coordinate Attention on Market-1501 and DukeMTMC-ReID under the same training protocol without ImageNet pretraining, metric-learning losses, or re-ranking.
- Found that Post and SharedBranch were generally more reliable than IndBranch, with AG-Post reaching 93.2% Rank-1 and 81.1% mAP on Market-1501 and CBAM-Post reaching 68.1% mAP on DukeMTMC-ReID.
- Analyzed the normalization sensitivity of CA-SharedBranch, showing that replacing BatchNorm with GroupNorm restored mAP from 28.6% to 80.7% on Market-1501 and from 21.9% to 68.1% on DukeMTMC-ReID.
- Accepted for presentation at ICDSAIA 2026.

DySample-ASFFLite YOLO11 for Lightweight Drone-Based Maritime Search-and-Rescue Object Detection

ICDSAIA 2026

ACCEPTED CONFERENCE PAPER

2026

- Developed a lightweight YOLO11s-based maritime search-and-rescue detector by replacing the two main top-down neck upsampling operators with DySample and adding an ASFFConcatLite adaptive feature-fusion block.
- Introduced a small/hard-category training-list rebalancing strategy to increase exposure to difficult classes such as life jackets and persons on boats in the SeaDronesSee dataset.
- Improved validation mAP50:95 from 0.1854 for the YOLO11s baseline to 0.2130 at epoch 30 while maintaining a lightweight configuration of 10.35M parameters and 24.8 GFLOPs.
- Observed improved qualitative recovery of small on-boat person detections compared with a larger YOLO11m baseline, supporting further ablation of dynamic upsampling and adaptive fusion.
- Accepted for presentation at ICDSAIA 2026.

An Efficient Human-Centric Violence Detection Framework with YOLOv11-AFNet and TemporalSA-MoViNet

PUBLISHED JOURNAL ARTICLE

Concurrency and Computation:
Practice and Experience

2026

- Developed a spatial-temporal video recognition pipeline combining YOLOv11-based person detection, AFNet feature learning, and TemporalSA-MoViNet for efficient violence recognition.
- Published by Wiley. DOI: 10.1002/cpe.70811.

Ghost-YOLOv11: A Lightweight and Efficient Deep Learning Model for Real-Time Mango Counting

PUBLISHED IEEE CONFERENCE PAPER

2026 IEEE 3rd International
Conference on Big Data Science and
Engineering (ICBDSE)

2026

- Developed a lightweight YOLOv11-based detector for real-time mango counting using GhostConv and C3k2Ghost modules, targeting efficient inference for resource-constrained vision systems.
- Achieved 1.85M parameters and 2.70 GFLOPs, with 0.773 MAE and 422.9 FPS on the reported test split.

HALE-YOLO: Integrating High-Resolution and Lossless Features for Efficient Aerial Tiny Person Detection

ACCEPTED CONFERENCE PAPER, PAPER ID: 207

HUMAN 2026

2026

- Worked on efficient aerial tiny person detection by integrating high-resolution and lossless feature representation for small-object detection.

Selected Projects

VNNETZERO – Digital Platform for Net Zero & Green Behavior Governance

FULL-STACK DEVELOPER

Vietnam National University, Hanoi

2026

- Built a production web platform using Next.js, TypeScript, Payload CMS, and PostgreSQL for publishing Net Zero research, courses, events, and sustainability content.
- Designed CMS schemas, dynamic routing, reusable content modules, multilingual architecture, and SEO workflows supporting Vietnamese, English, Chinese, Korean, and Japanese content.
- Containerized and deployed the platform with Docker and a reverse proxy on a Linux VPS, including domain, HTTPS, database, and production configuration.

SAYNTAX – Voice-Driven Coding for Accessible Web Development

PROJECT LEAD

GenAct “STEAM for ALL”

2026

- Led an inclusive technology initiative enabling people with disabilities to explore web development through voice-driven coding and accessible interaction methods.
- Designed the technical workshop workflow, participant learning experience, pre/post evaluation, and community ambassador model for pilot implementation.
- Selected as a Top 10 project in the GenAct “STEAM for ALL: Technology for an Inclusive World” challenge, receiving seed funding to implement and evaluate the pilot.

School-Violence Detection System from CCTV Using Video, Deep Learning, and Attention

UNIVERSITY-LEVEL RESEARCH PROJECT

FPT University

08/2025 - 08/2026

- Built an end-to-end video analysis pipeline using person detection, feature extraction, attention, and temporal modeling for CCTV footage.
- Contributed to dataset construction and cleaning, model implementation, experiment design, evaluation, reporting, and manuscript revision.

Person Re-Identification on OSNet via MDHA & Dynamic Graph Convolution

RESEARCH MEMBER

FPT University

08/2025

- Enhanced OSNet with MDHA and Dynamic Graph Convolution; evaluated performance on Market-1501 and DukeMTMC and conducted ablation studies.
- Contributed to module design, experiments, and technical writing; related multi-camera tracking work was presented at Research Festival 2025.

Experience

FPT Software - AI Valley

AI INTERN

Can Tho, Vietnam

01/2026 - 04/2026

- Contributed to production-oriented AI workflows for enterprise use cases, including preprocessing, model training and optimization, inference setup, and validation.
- Participated in deployment and integration workflows using ONNX, Docker, Linux environments, and REST APIs.

FPTU AI & Robotics Challenge 2025 / Vietnam Open Robotics Challenge 2024

TECHNICAL MENTOR

Can Tho, Vietnam

10/2024 - 08/2025

- Supported teams with AI problem solving, ESP32, sensors, robotics hardware, connectivity, and on-site troubleshooting during competition deployment.

Education

- Relevant courses: Computer Vision, Deep Learning, Machine Learning, Data Mining, Linear Algebra, Probability & Statistics, Python for AI.

Awards & Certifications

08/2025	Consolation Prize , Research Festival 2025 - online multi-camera people tracking with MDHA Attention and DGC Convolution	<i>FPT University</i>
04/2026	Specialization , Natural Language Processing - DeepLearning.AI	<i>Coursera</i>
12/2025	Certificate , Introduction to Containers w/ Docker, Kubernetes & OpenShift	<i>Coursera / IBM</i>
06/2026	Certificate , Practical Quantum Computing with IBM Qiskit for Beginners	<i>Coursera</i>
03/2025	2nd Prize , “Var-Code” programming and problem-solving competition	<i>FPT University</i>

Leadership & Activities

- Managed club operations and technical activities; designed AI/ML training content, guided mini-projects, and coordinated AI/robotics workshops and competition support.

- Coordinated youth programs related to sustainability, green jobs, and capacity building, including cross-functional planning across people, communications, and event operations.
- Contributed to initiatives that delivered 30+ capacity-building workshops and directly benefited approximately 1,400 youth.